

# MUZAMMIL HAIDER

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## EDUCATION

**BS in Artificial Intelligence** — National University of Modern Languages, Islamabad | CGPA: 3.60/4.0 Feb 2023 – Present

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL, JavaScript

**AI / ML:** TensorFlow, PyTorch, Scikit-learn, LangChain, CrewAI, AutoGen

**Specialties:** LLMs, RAG Systems, NLP, Computer Vision, OSINT, Web Automation

**Automation:** Playwright, Selenium, Undetected-Chromium, Obscura, anti-detection & fingerprint bypassing

**Frontend:** Next.js, React, Tailwind CSS

**Backend & Tools:** FastAPI, Django, Streamlit, Docker, Supabase, Firebase, Render, n8n, Make

**Data:** Pandas, NumPy, OpenCV, NLTK

## EXPERIENCE

**Automation Engineer** — NACTA (National Counter Terrorism Authority) Jan 2026 – Present

- Built end-to-end automation tools for large-scale data gathering and scraping from public/semi-public sources in support of investigative workflows
- Developed stealth browser automation pipelines using Undetected-Chromium and Obscura to bypass rate limiting, CAPTCHAs, and bot-detection on high-value targets
- Built OSINT tooling to systematically collect, structure, and correlate investigative data across platforms; distributed scraping with rotating proxies and fingerprint spoofing

**Data Scientist** — 10Pearls Pakistan (Shine Program) Mar 2026 – May 2026

- Built Islamabad AQI Predictor: 24-hour ML forecasting (0.89 R<sup>2</sup>) using XGBoost, LightGBM, and Random Forest ensemble with full MLOps pipeline — GitHub Actions CI/CD, Hopsworks Feature Store, daily retraining

**AI Intern** — Bewhoop Jun 2025 – Aug 2025

- Designed AI-powered outreach automation agent (LangChain, FastAPI, Supabase); built lead generation and follow-up workflows with n8n and Make integrated with LLM-based decision logic

## PROJECTS

**TikTok Scraper (API-free)** Playwright · Undetected-Chromium · Obscura · FastAPI 2025

- Fully API-free TikTok scraping system extracting posts, profiles, hashtag feeds, and engagement metrics with zero reliance on official or unofficial APIs
- Bypassed TikTok's aggressive rate limiting and bot detection using Obscura for fingerprint randomization — canvas noise, WebGL spoofing, user-agent rotation, and behavioral mimicry (mouse patterns, scroll simulation, randomized delays)
- Distributed task queue with proxy rotation and automatic session recovery; FastAPI interface for configurable collection jobs with JSON / CSV / Supabase output

**AI SQL Analytics Agent** LangGraph · Gemini 2.0 · FastAPI · Next.js · Tailwind CSS · Supabase Nov 2025

- Autonomous multi-agent system translating natural language to SQL via LangGraph orchestration; 8 parallel agents handle insight detection, anomaly detection, correlation, trend forecasting, and KPI tracking simultaneously
- Generates McKinsey-grade PDF reports with 3+ visualizations in ~2 minutes (85–90% token reduction vs. naive LLM approaches); full-stack Next.js 16 + Tailwind CSS frontend with real-time dashboards

**Agentic Data Analyst & Report Generator** LangChain · Gemini 2.0 · Supabase · Streamlit Nov 2025

- Chain-of-thought SQL agent with multi-step reasoning; 3-tier DB fallback (PostgreSQL → REST API → Pandas); prompt engineering with schema injection and automated duplicate detection across 6-section reports

**Alif — AI Tutor App** TensorFlow · PyTorch · NLTK · FastAPI · Next.js · Tailwind CSS Dec 2024 – Present

- Full-stack intelligent tutoring system with adaptive learning paths (TensorFlow/PyTorch), NLP pipelines, voice interaction, and responsive Next.js + Tailwind CSS frontend

## CERTIFICATIONS & HACKATHONS

- Machine Learning Specialization — DeepLearning.AI & Stanford University (Coursera), 2024
- NCEAC-HEC Generative AI Training Program, Cohort 1 — Pak Angels and Partners, Oct 2025
- Hackathon Participant — COMPECC NUST 2024 & 2025 | VISIO SPARK COMSATS 2024